Thesis for Doctor Degree

Role of CD150 expression in Epstein-Barr virus-transformed B cells

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(Anatomy)

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A thesis submitted to the Graduated School of Inje University in partial fulfillment of the requirements for the degree of Doctor of Philosophy in Medicine

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국문초록

엡스타인바 바이러스 변형 B세포에서 발현된 CD150의 기능

서 일 경
(해부학전공)
(지도교수: 허대영)
의학과
인제대학교 대학원

목적 : CD150, 림프구 활성신호 물질(SLAM)은 면역세포와 일부 암에서 발현하고 있는 세포표면 수용체이다. 면역세포에서 CD150을 자극하면 세포증식반응과 사이토카인 분비가 유도된다. 하지만, 엡스타인바 바이러스 감염 B세포에서의 CD150의 기능은 알려진 바가 아직 없다. 본 연구의 목적은 병적인 과정의 결과로 여겨지는 엡스타인바 바이러스에 의한 B세포의 변형과정에서 CD150의 발현 변화를 확인하고, 나아가서 엡스타인바 바이러스 변형 B 세포에서 발현이 증가된 CD150 물질의 기능적 역할과 작용 기전을 밝히고자 하였다.

방법 : CD150 물질 자극 세포모델로 엡스타인바 바이러스 변형 B세포를 확립하였다. 엡스타인바 바이러스 감염 B세포에서 CD150 발현을 확인하기 위하여 공초점현미경, 유세포분석, 통상적인 역전사중합효소반응, 실시간 정량적 중합효소반응을 실험하였다. 엡스타인바 바이러스 변형 B세포에서 CD150의 기능적 역할과 작용 신호전달 기전을 밝히기 위하여 부착분자 발현, 표현형 변화, 사이토카인 생산 유무를 확인하기 위해 형태학적 관찰, 유세포 분석, 증식반응 분석, 실시간 중합효소반응과 효소결합면역흡착측정법, 웨스턴 블롯법, 신호전달 억제제분석방법을 사용하였다.

결과 : 엡스타인바 바이러스 감염은 B세포의 CD150 발현을 증가시켰다. CD150 발현이 증가된 엡스타인바 변형 B 세포에 이 물질을 자극하면 부착분자의 발현이 증가되고 세포무리 형성이 증가되었다. 이에 더하여 엡스타인바 변형 B 세포에 항-CD150 항체를 처리하면 세포증식반응이 증가하고, 표현형 변화와 사이토카인 생산이 유도되었다. CD150 매개 신호전달 기전은 PI3K/AKT와 SRC/ERK/JNK 효소의 인산화를 통해서 일어남을 웨스턴 블롯법과 신호전달 억제제 분석결과를 통해 확인할 수 있었다.

결론 : 전체적으로 볼 때, 본 연구결과는 CD150 자극에 의한 엡스타인바 바이러스 변형 B세포 반응에 대한 새로운 개념을 보여주는 것으로, 엡스타인바 바이러스 변형 B세포가 비정상적인 조절 사이토카인 분비의 표적세포가 될 수도 있으며, 국소적 그리고 전체적 사이토카인 네트워크에 잠재적 활동 세포가 될 수도 있음을 보여줍니다. 엡스타인바 바이러스 변형 B세포와 엡스타인바 바이러스 양성 림프종에서 발암과 사이토카인 생산에 관여하는 CD150의 잠재적인 기능에 대한 명확한 이해는 추가적인 연구가 필요할 것으로 생각됩니다.

중심단어 : CD150, 엡스타인바 바이러스 (EBV), 부착분자, 사이토카인 생산, 신호전달 기전

ABSTRACT

Role of CD150 expression in Epstein-Barr virus-transformed B cells

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Objective : CD150, also called signaling lymphocyte activation molecule (SLAM), is a cell surface receptor expressed on immune cells and some tumors. Stimulating CD150 on immune cells induces cell proliferation and cytokine production. However, the function of CD150 in Epstein-Barr virus (EBV)-infected B cells is not fully understood. The objective of this study was to assess the changes in CD150 expression as a pathologic consequence of EBV-induced B cell transformation and examine the functional role and underlying mechanism of an increased in the number of CD150 molecules in EBV-transformed B cells.

Methods : EBV transformed B cells were established as a cellular model of CD150-stimulated cells. To evaluate CD150 expression in EBV-infected B cells, confocal microscopy, flow cytometry analysis, reverse transcription-polymerase chain reaction (RT-PCR), and real time quantitative PCR were performed. To verify the functional role and underlying signaling mechanism of CD150 in EBV-transformed B cells, adhesion molecule expression, phenotype changes, and cytokine production were assessed using a morphological investigation, flow cytometry, a proliferation assay, real-time PCR, enzyme linked immunosorbent assays, western blotting, and signaling inhibition assays.

Results : EBV infection increased CD150 expression on B cells. The increased CD150 stimulation induced adhesion molecule expression and clump formation in EBV-transformed B cells. Additionally, treatment with anti-CD150 antibodies increased cell proliferation, and induced phenotype changes and cytokine production in EBV-transformed B cells. Western blots and the signaling inhibition assays showed that the CD150-mediated signal mechanism operated through the phosphorylation of the PI3K/AKT and SRC/ERK/JNK kinases.

Conclusion: Taken as a whole, these results provide a novel insight into the mechanism of CD150 stimulation in EBV-transformed B cells, and suggest that EBV-transformed B cells should be considered both as targets for an unusual regulatory cytokine and potentially active participants in local and systemic cytokine networks. More studies are needed to better understand the potential roles of CD150 in carcinogenesis and cytokine production in EBV-transformed B cells and EBV^{+} lymphoma.

Key words : CD150, Epstein-Barr virus (EBV), adhesion molecule, cytokine production, signaling mechanism

I. Introduction

CD150 (signaling lymphocyte activation molecule; SLAM) is a type I trans-membrane glycoprotein belonging to the CD2/CD150 family of the immunoglobulin (Ig) superfamily of proteins [1, 2]. High levels of CD150 expression are frequently associated with B cell malignancies and are implicated in immunodeficiency X-linked lymphoproliferative disease (XLP) [3, 4]. Engaging CD150 with an anti-CD150 mAb enhances cell proliferation and cytokine production and stimulates some and inhibits other important regulatory signals [1, 5, 6]. CD150 also acts as a receptor for cellular entry for several morbilliviruses, including the measles virus (MV) [7]. CD150-expressing T cells in the human immune system are important target cells for wild-type MV [8], and infection of those cells can contribute to immune suppression by reducing the expression of soluble IL-2 receptor, interferon-gamma (IFN-ɣ), IL-1 beta, and tumor necrosis factor alpha [9, 10].

$CD4^{+}$ helper T cells are critical to germinal center responses, and their activation depends on T cell-dendritic cell (DCs) interactions [11, 12]. Subsequent cognate interactions between activated B cells and T cells induce germinal center formation. Mice deficient in signaling lymphocyte activation molecule (CD150)-associated protein (SAP) have a marked defect in B-cell proliferation and germinal-center formation [13]. However, the relationship between activated B cells and DCs associated with SLAM still requires investigation. Despite several studies showing that CD150 has various immunological functions in cytokine production and intracellular signaling in T cells and other cell types, very little is known about the roles and mechanisms that regulate CD150-induced cytokine secretion in virus-infected

B cells or B-cell malignancies.

CD54 (intercellular adhesion molecule-1) is one of the most concentrically observed cell adhesion molecules, and a membrane protein expressed not only in endothelial cells and several lymphocytes, but also in diverse cancer cells [14]. Lymphocyte function-associated antigen-1 and its major ligand, CD54, play a role in tumor growth and metastasis [15]. CD22 is another cell adhesion molecule, and a B lymphocyte specific glycoprotein that first appears as B cells are developing, disappearing when B cells differentiate into plasma cells in the germinal center [16]. SAP co-localizes and interacts with CD22 [17]. However, the actions of adhesion molecules induced by CD150 stimulation of Epstein-Barr virus (EBV)-transformed B cells remain poorly understood.

In this study, EBV-transformed B cells were used as a cell model system to clarify the signaling mechanisms by which adhesion molecules and cytokines are produced by CD150-stimulation.

Ⅱ. Objectives

CD150 (SLAM) is a cell surface receptor expressed on immune cells and some tumors. Stimulation of CD150 on immune cells induces cell proliferation and cytokine production. However, the function of CD150 in Epstein-Barr virus (EBV)-infected B cells is not fully understood. The objective of this study was to assess changes in CD150 expression as a pathologic consequence of EBV-induced B cell transformation and examine the functional role and underlying mechanism of the increased number of CD150 molecules found in EBV-transformed B cells.

Ⅲ. Materials and Methods

A. Generation of EBV-transformed B cells and culture conditions

EBV supernatant stock was prepared from an EBV B95-8 marmoset cell line (a free gift from Dr. Han, Korea National Institute of Health, Seoul, Korea). Peripheral blood mononuclear cells (PBMCs) were isolated from the whole blood of two healthy human volunteers (informed consent was obtained from each participant) using Ficoll-Paque (Amersham Life Science, Buckinghamshire, United Kingdom) gradient centrifugation. B cells were purified from the PBMCs using a B cell isolation kit (Miltenyi Biotec, Bergisch Gladbach, Germany). To establish EBV-infected B cells, purified B cells were added to EBV stock supernatant. After a 2 hours incubation at 37°C culture medium containing RPMI-1640 (HyClone, Logan, UT, USA) was added (final cell concentration; 5×10^5 cells/ml). Cells were incubated in RPMI-1640 medium (HyClone) replenished with 10% fetal bovine serum (Tissue Culture Biologicals, Tulare, CA, USA), 100 IU/ml penicillin (HyClone), and 100 mg/ml streptomycin (HyClone) at 37℃ in a 5% CO_2 humidified atmosphere. The cultures were maintained for 2 to 4 weeks until cell-clumps of EBV-infected cells were visible and the medium turned yellow. After the cells were grown to sub-confluence, they were treated with different concentrations of anti-human CD150 antibodies (Affinity Bioreagents, Golden, CO, USA) or UPC-10 (isotype control, Sigma-Aldrich, St. Louis, MO, USA) for 0, 1, 3, 6, 9, 12, 24, and 48 hours. This study was approved by the Institutional Bioethics Review Board at the Inje University Busan Paik Hospitals and all the donors gave informed consent for their participation.

B. Reagents and antibodies

Purified anti-human CD150 Ab was purchased from Affinity Bioreagents (Golden, CO, USA). UPC-10 (mouse IgG2a) was purchased from Sigma-Aldrich (St. Louis, MO, USA) and used as an isotype control. Fluorescein isothiocyanate (FITC)-conjugated anti-human CD20, CD22, CD38, and CD80 antibodies, and phycoerythrin (PE)-conjugated anti-human CD150, CD54, IL-10, and IgD antibodies were obtained from BD Biosciences (San Jose, California, USA). Anti-phospho-PI3K p85 (p-PI3K), anti-PI3K p85 (PI3K), anti-phospho-Akt (p-AKT), anti-Akt (AKT), anti-phospho-SRC (p-SRC, Y416), anti-SRC, and anti-β-actin antibodies were purchased from Cell Signaling Technology (CST, Danvers, MA, USA) for use as primary antibodies. Horseradish peroxidase-conjugated secondary antibodies were purchased from Amersham Bioscience. SP600125 (Calbiochem, La Jolla, CA, USA), a selective, reversible ATP-competitive inhibitor of JNK; SB203580 (Calbiochem), a specific inhibitor of ERK1/2; and LY294002 (Calbiochem), a specific inhibitor of PI3K, were used to evaluate the signaling mechanism that induced by CD150 stimulation on EBV-transformed B cells.

C. Flow cytometry analysis

A flow cytometry analysis was performed to evaluate the expression of CD150, CD54, CD22, IL-10, IgD, CD38, and CD80. Shortly, the cells were washed thrice with cold phosphate buffered saline (PBS, Sigma-Aldrich). Then, the cells were retained on ice for 40 min with the FITC- or PE-conjugated mouse anti-human primary Abs described previously. All samples were evaluated using a flow cytometer (FACSCalibur, BD Biosciences), and appraised data were analysed using the CELLQUEST program (BD Bioscience).

D. Confocal microscopic image analysis

Confocal microscopic images were analyzed to clarify the expression of CD150 and CD20 molecules. Shortly, the cells were washed thrice with cold PBS (Sigma-Aldrich, St. Louis, MO, USA), and then incubated on ice for 30 min with FITC-conjugated mouse anti-human CD150 and PE-conjugated CD20 Abs. To detect the intracellular molecules, cells were pre-treated with permeabilization solution (PBS buffer containing 0.1% saponin). After being washed twice with PBS, all the prepared cells were put on microscopic slides using fluorescent mounting medium (DakoCytomation, Glostrup, Denmark) and placed under cover-slips. Cells stained with fluorescence were observed by confocal microscopy (510 META model, Carl Zeiss, Jena, Germany) at ×400 magnification, and acquired images were analysed using Confocal Microscopy Software Release 3.0 (Carl Zeiss).

E. Reverse transcription polymerase chain reaction (RT-PCR) and real-time quantitative PCR

Total RNA was acquired with an RNeasy Mini kit (Qiagen, Hilden, Germany). Isolated RNA was transcribed into cDNA using oligo-nucleotide primers and reverse transcriptase enzyme. To investigate the level of various cytokines, PCR amplification was performed using specific primer sets (Bioneer, Daejeon, South Korea) for CD150, IL-1α, IL-1β, IL-4, IL-5, IL-6, IL-8, IL-10, IL-12p35, IL-12p40, IL-17α, GM-CSF, and β-actin (Table 1). PCR was performed using Prime Taq Premix (GeNet Bio, Daejeon, South Korea). Quantitative real-time RT-PCR was carried out with a SYBR Green kit (Takara, Shiga, Japan), iCycler thermal real-time PCR system (BioRad, Hercules, CA, USA), and the same primer sets used in the conventional RT-PCR (Table 1). The relative gene expression compared with control cells was evaluated using the iCycle iQ real-time detection system software's built-in algorithm (BioRad) and an adaptive baseline to analyse the threshold cycle (Ct). mRNA fold induction values were estimated according to the following equations: $\triangle Ct = Ct_{\text{target}} - Ct_{\beta\text{-actin}}$, $\triangle(\triangle Ct) = (\triangle Ct_{\text{stimulated}} - \triangle Ct_{\text{control}})$, mRNA fold change = $2^{-\triangle(\triangle Ct)}$. Experiments were practiced in quadruplicate, and data are articulated as means ± standard deviation (SD).

Table 1. Summary of oligonucleotide primers sequences used for RT-PCR and real-time PCR.

Target	Primers, 5' → 3'		Product size
	Sense	Anti-sense	
CD150	TAT CTA CAT CTG CAC CGT GAG C	TCC TGA GCT GGG AAG GAG T	288 bp
IL-1A	CTG CAT GGA TCA ATC TGT	CCC ATG TCA AAT TTC ACT GC	369 bp
IL-1B	CAG CTA CGA ATC TCC GAC CAC	GGC AGG GAA CCA GCA TCT TC	100 bp
IL-4	ATG GGT CTC ACC TCC CAA CTG CTT	TTT CCA ACG TAC TCT GGT TGG C	355 bp
IL-5	TCT GAG GAT TCC TGT TCC TG	TTA TCC ACT CGG TGT TCA TT	248 bp
IL-6	GTG TTG CCT GCT GCC TTC CCT G	CTC TAG GTA TAC CTC AAA CTC CAA	321 bp
IL-8	ATG ACT TCC AAG CTG GCC GTG GCT	TCT CAG CCC TCT TCA AAA ACT TCT C	292 bp
IL-10	CTG AGA ACC AAG ACC CAG ACA TCA AGG	GTC AGC TAT CCC AGA GCC CCA GAT CCG	327 bp
IL-12A	CTT CAC CAC TCC CAA AAC CTG	AGC TCA TCA CTC TAT CAA TAG	532 bp
IL-12B	CAT TCG CTC CTG CTG CTT CAC	TAC TCC TTG TTG TCC CCT CTG	266 bp
IL-17	ATG ACT CCT GGG AAG ACC TCA TTG	TTA GGC CAC ATG GTG GAC AAT CGG	156 bp
GM-CSF	ATG TGG CTG CAG AGC CTG CTG C	CTG GCT CCC AGC AGT CAA AGG G	424 bp
β-actin	ATC CAC GAA ACT ACC TTC AA	ATC CAC ACG GAG TAC TTG C	200 bp

F. Investigation of morphologic changes

EBV-t B cells were alloted at a density of 2 × 10^5/well in six-well plates and incubated for three days, and then they were treated with anti-human CD150 Ab, or UPC-10 Ab (as an isotype control) at the same concentration. Cells were then observed with an inverted microscope digital imaging system (CELENA®S Digital Fluorescence Imaging System; Logos Biosystems, Anyang-si, Gyeonggi-do, South Korea).

G. Proliferation measurement by AlamarBlue assay

Anti-CD150 Ab (100, 200, 500 ng/ml, and 1 ㎍/ml in PBS) and UPC-10 Ab (isotype control, Sigma-Aldrich, 1 ㎍/ml in PBS) were separately placed in 96-well culture plates (0.1 ml/well; washed with PBS before use) after 12 hours incubation at 4°C. EBV-t B cells (5×10^5 cells/well, 200 ㎕) were incubated at 37°C in a plate coated with antibodies for 48 hours. These cell proliferations were evaluated with an AlmarBlue assay kit (Serotec, Raleigh, NC, USA). Briefly, AlmarBlue solution (10% by total volume) was added to each well, and after 8 hours incubation the relative fluorescence value was assessed with a Synergy HT fluorometer (Bio-Tek Instruments, Winnoski, VT; excitation, 530 nm and emission, 590 nm).

H. Quantification of human cytokines by ELISA

Culture supernatant from CD150-stimulated EBV-transformed B cells or EBV^+ lymphoma cell lines was concentrated using an Amicon Ultra-15 (Millipore). IL-1α, IL-1β, IL-4, IL-5, IL-6, IL-8, IL-10, IL-12, IL-17α, and GM-CSF concentrations in the 10-fold concentrated cell-free culture supernatants were determined by multi-cytokine enzyme linked immunosorbent assay ELISA (Multi-Analyte ELISArray; SABiosciences, Frederick, ML, USA) and also quantified by single cytokine ELISA (Single Analyte ELISArray;

SABiosciences). The data are expressed as the average number of biological replicates ± standard deviation (SD).

I. Western blotting

EBV-transformed B cells were harvested and washed twice with PBS (Sigma-Aldrich), and then cell extracts were prepared using RIPA buffer (Elpis Biotech, Daejeon, Korea). RIPA buffer was composed of a protease inhibitor cocktail (bestatin hydrochloride, AEBSF, aprotinin, EDTA, E-64, and leupeptin hemisulfate salt; Sigma-Aldrich). To facilitate the measurement of phosphorylation, a set of phosphatase inhibitors (sodium molybdate, sodium orthovanadate, sodium tartrate, and imidazole; Sigma-Aldrich) was added to the RIPA buffer. Protein concentration was evaluated using a BCA assay kit (Pierce, Rockford, IL, USA). Western blot analyses were conducted as follows. Equal amounts of proteins (10 μg/sample) were segregated by electrophoresis on SDS-polyacrylamide gels, and then segregated proteins transferred to polyvinylidene difluoride membranes (Amersham Life Science, Braunschweig, Germany) by immuno-blotting. The membranes were then incubated overnight at 4°C in a PBS solution supplemented with 5% non-fat dry milk. Chemiluminescence was visualized using an ECL kit (Amersham Life Science) and Gel DOC system (Fujifilm, Tokyo, Japan). The following primary antibodies were used: β-actin, phospho-PI3K p85, PI3K p85, phospho-Akt, Akt, phospho-SRC, and SRC from Cell Signaling Technology (Danvers, MA, USA). Each experiment was repeated at least three times.

J. Evaluation of individual MAPK inhibitors following CD150 stimulation

To evaluate the mechanism of the CD150-stimulated cellular response in EBV-t B cells, these cells were pre-incubated with SP600125 (Calbiochem, La Jolla, CA, USA), a selective, reversible ATP-competitive inhibitor for JNK;

SB203580 (Calbiochem), a specific inhibitor for ERK1/2; or LY294002 (Calbiochem), a specific inhibitor for PI3K, for 2 hours before stimulation with anti-CD150 Ab. Cells were washed to remove all chemicals before the anti-CD150 Ab treatment. EBV-t B cells (2×10^5 cell/well, 200 $\mu\ell$) were incubated in plates coated with anti-CD150 Ab under the same conditions used in the previous experiments. Phenotype changes were then evaluated using flow cytometer (FACSCalibur, BD Biosciences, San Diego, CA, USA), and data were analyzed under the CELLQUEST program (BD Bioscience).

K. Statistical analysis

All experrimental data are showed as mean ± SD, and each value represents at least two different experiments. Comparisons between individual data points were analysed by one-way analysis of variance (ANOVA). A p-value < 0.05 was considered statistically significant.

Ⅳ. Results

A. EBV infection increased CD150 expression on B cells

To determine CD150 expression patterns, EBV-infected B cells were observed with conforcal microscopy and flow cytometry every week for 4 weeks during the EBV-induced transformation process. CD150 was minimally expressed on the surface and in the cytoplasm of $CD20^+$ resting B cells. Expression of CD150 was slightly increased one week following infection and rose quickly thereafter. By three weeks after EBV infection, most of the EBV-transformed B cells strongly expressed CD150 molecules on the cell surface and in the cytoplasm (Fig. 1 and Fig. 2). CD150 mRNA had also increased slightly by one week following EBV infection and had likewise increased significantly by 3 weeks after EBV infection (Fig. 3).

B. CD150 stimulation induced clump formation and increased the expression of adhesion molecules in EBV-t B cells

To clarify the role of CD150 induced by EBV infection, EBV-t B cells were treated with anti-CD150 Ab. First, changes in morphology were investigated, which showed that EBV-t B cells aggregated and formed large clumps after treatment with anti-CD150 Ab (Fig. 4). Next, the cause of the clump formation was investigated by testing for an increase in adhesion molecules. As expected, anti-CD150 Ab treatment increased the expression of adhesion molecules (both CD54 and CD22) in EBV-t B cells (Fig. 5).

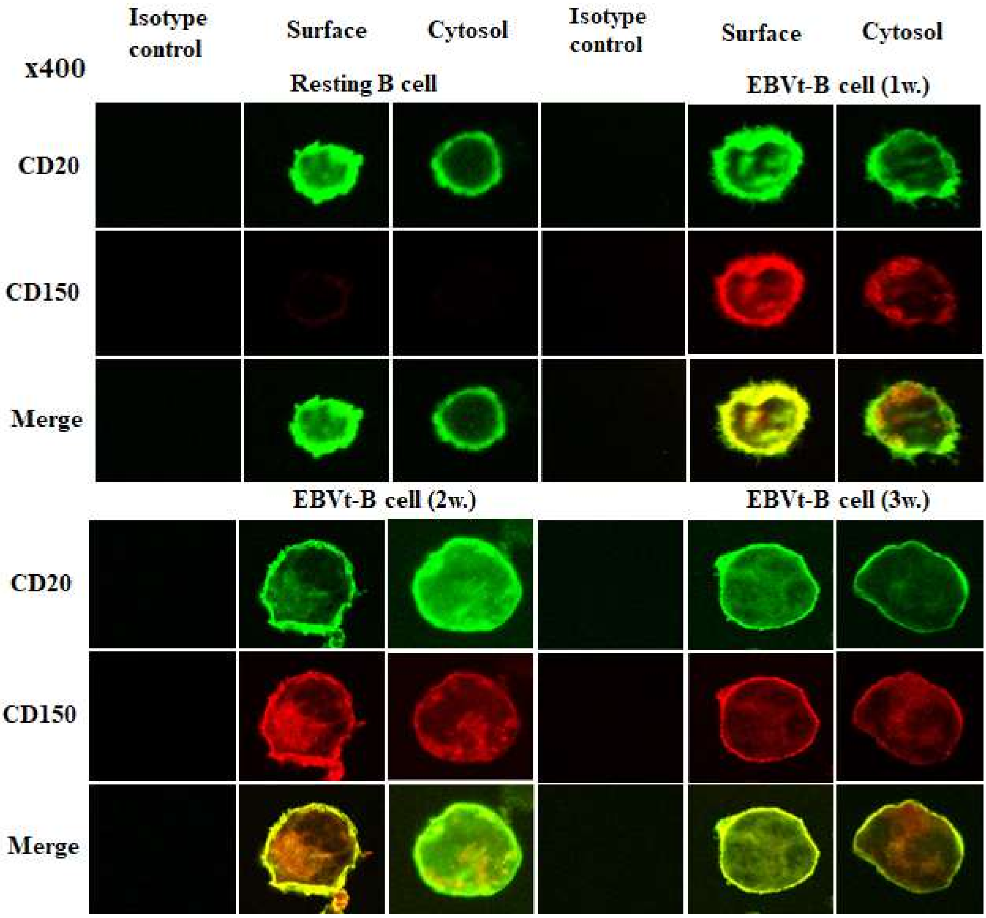

Figure 1. Confocal microscope images of CD150 expression in EBV-infected B cells at different times. Confocal microscope images showing the expression of CD20 (green) and CD150 (red). At 1 week, EBV-infected B cells (CD20 positive; green fluorescence) clearly expressed CD150 on both the cell membrane and cytoplasm. After 2 weeks of EBV infection, EBV-t B cells highly expressed CD150 molecules. EBVt-B cell, Epstein-Barr virus-transformed cells; x400 magnification.

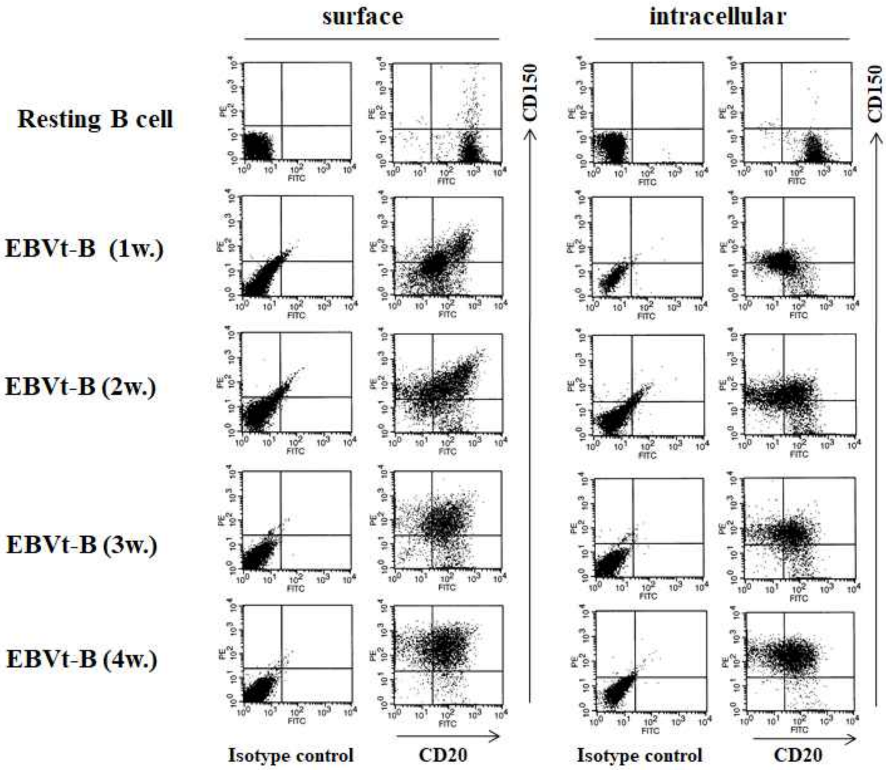

Figure 2. Expression of CD150 protein during B cell transformation by EBV. Cells were harvested (10^6 cells/200$\mu\ell$) and stained with a PE-conjugated anti-CD150 Ab and FITC-conjugated anti-CD20 Ab for FACS analysis. The scatter-dot plots on the left exhibit cell membrane expression, and those on the right exhibit intracellular molecule expression. Results represent three independent experiments. EBV-t B, Epstein-Barr virus-transformed B cells; w., week(s).

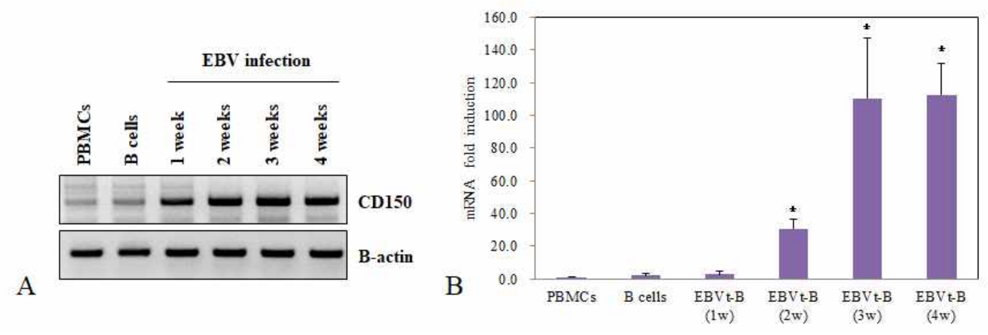

Figure 3. Expression of CD150 mRNA during B cell transformation by EBV. (A) Representative photomicrographs show the expression of CD150 mRNA at different times during transformation by EBV infection. (B) Graph shows the relative gene expression compared with PBMCs. The results represent four independent experiments. Data represent the mean ±SD of three experiments. *$p<0.01$, EBVt-B groups versus B cells (normal control) groups (student's t test). PBMCs, peripheral blood mononuclear cells ;B-actin, beta-actin; EBV-t B, Epstein Barr virus-transformed B cells; w, week(s).

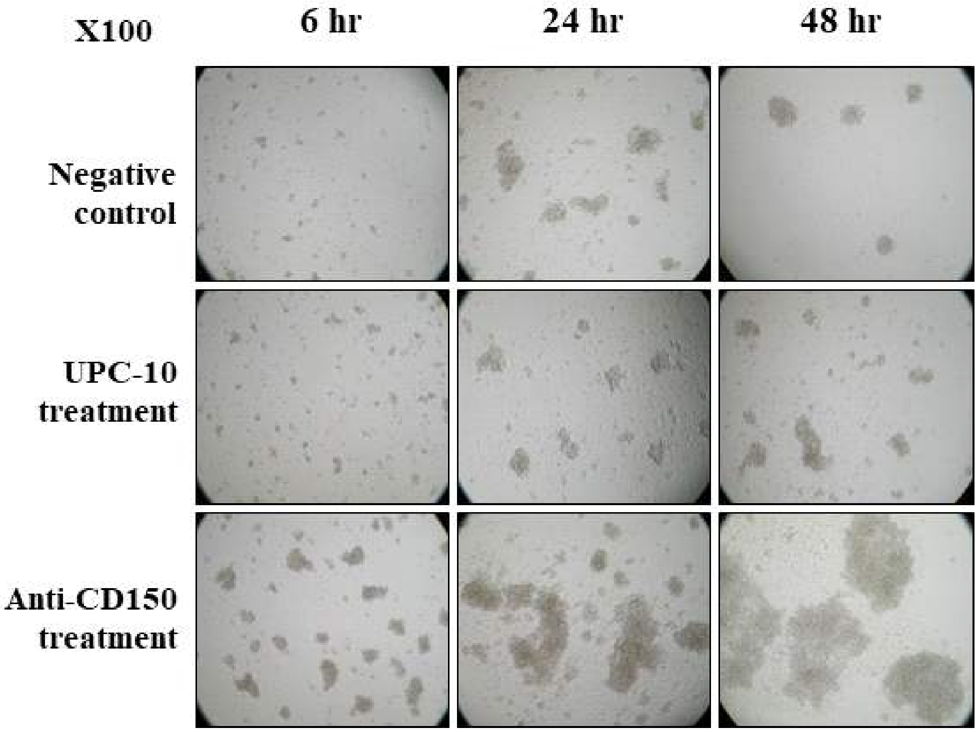

Figure 4. Morphological changes in EBV-t B cells after anti-CD150 Ab treatment. Representative photomicrographs show morphological changes (clump formation) after isotype control (UPC-10) Ab or anti-CD150 Ab treatment (x100 magnification). Results represent five independent experiments. Hr, hours.

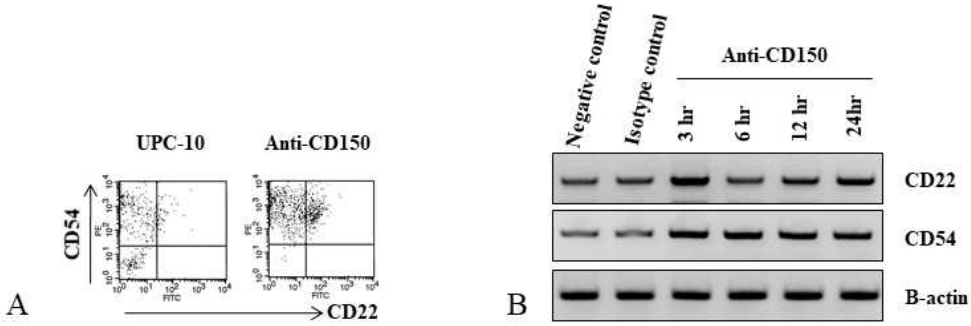

Figure 5. Changes in the expression of adhesion molecules in EBV-t B cells after anti-CD150 Ab treatment. EBV-t B cells were incubated with anti-CD150 or UPC-10 (isotype control) antibody for 24 h. (A) Cells were harvested and then stained with FITC-conjugated anti-CD22 Ab and PE-conjugated anti-CD54 Ab for FACS analysis. Scatter-grams represent surface expression. (B) Cells were harvested, total RNA isolated, reverse transcribed into cDNA, and then PCR amplifications were performed as described previously. Results represent three independent experiments. B-actin, beta actin (used as a control); hr, hours; anti-CD150, anti-CD150 antibody treatment.

C. CD150 stimulation increased the proliferation of EBV-t B cells

To evaluate the effect of CD150 stimulation on EBV-transformed B cells, the AlamaBlue proliferation assay was performed. CD150 stimulation with anti-CD150 antibodies increased the proliferation of EBV-transformed B cells in a dose-dependent manner compared with untreated negative control and isotype control (UPC-10) cells. These increases were statistically significant above concentration of 200 ng/ml (**p<0.01 and *p<0.05, Fig. 6).

D. CD150 stimulation induced phenotype changes and cytokine production in EBV-t B cells

Next, I investigated whether the acceleration of cell proliferation was related to phenotype changes and cytokine production. EBV-transformed B cells were stimulated with anti-CD150 Ab or mock stimulated with the UPC-10 isotype control, followed by cross-linking with secondary Ab. CD150 stimulation induced phenotype changes indicating B cell activation (IL-10, IgD, CD38, CD80) in a time-dependent manner (Fig. 7). Next, changes in the expression of inflammatory cytokines were evaluated with cytokine ELISA assay kits and quantitative real-time RT-PCR. The mRNA of several inflammatory cytokines (IL-1α, IL-1β, IL-4, IL-8, IL-5, IL-6, IL-10, IL-12, IL-17α, and GM-CSF) increased in EBV-transformed B cells for 3 to 12 h after stimulation with anti-CD150 mAb (Fig. 8). Protein levels of these cytokines also increased dramatically 24 h after CD150 stimulation, compared with the levels secreted by EBV-transformed B cells after mock stimulation with UPC-10 (**p<0.01 and *p<0.05). Notably, the production of IL-1α and IL-1β following CD150 stimulation was much higher than that of the other cytokines detected at the protein level. Although the mRNA levels of IL-4, IL-6, and IL-10 increased slightly, protein production was not increased significantly compared with untreated EBV-transformed B cells and those treated with the isotype control (Fig. 9).

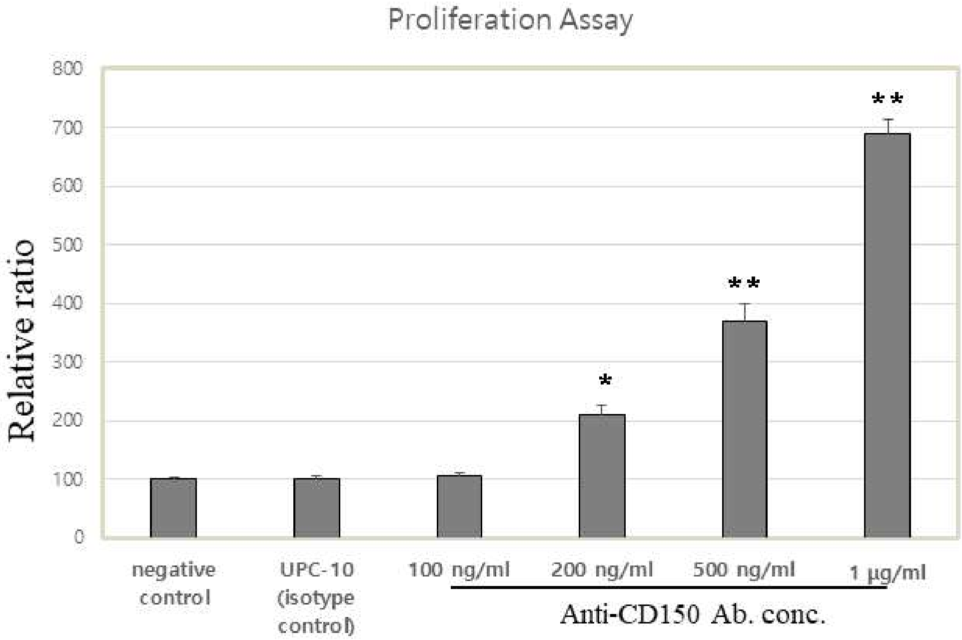

Figure 6. Proliferation assay in EBV-t B cells treated with anti-CD150 Ab. EBV-t B cells were incubated with anti-CD150 or UPC-10 (isotype control) antibody for 48 h. After 7 hours incubation, AlamarBlue solution was added (10% by volume) to each well, and then the relative fluorescence was measured with a fluorometer. Data represent the mean ±SD of three experiments. *p <0.05 or **p<0.01, CD150-stimulated groups versus UPC-10 (isotype control) groups (student's t test).

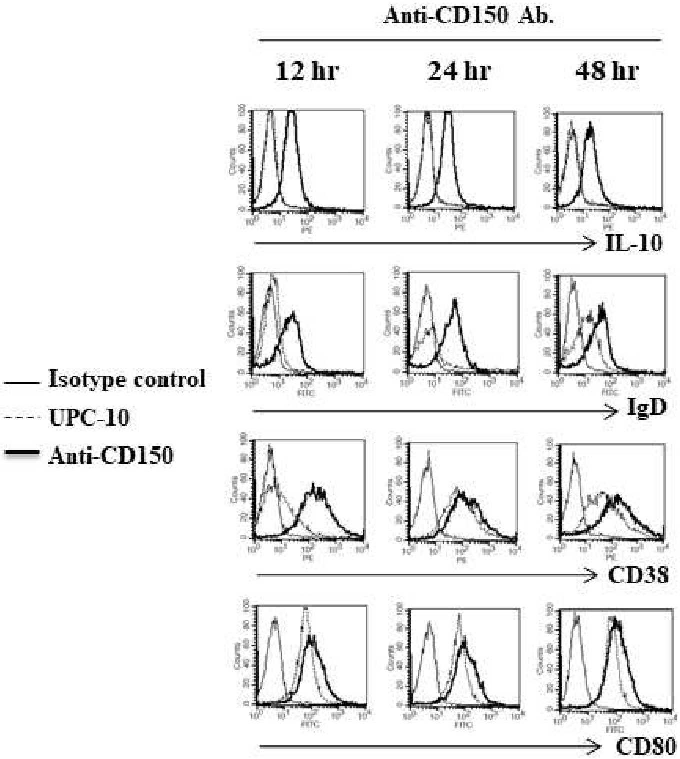

Figure 7. Phenotype changes in EBV-t B cells after anti-CD150 antibody treatment. EBV-t B cells were incubated with anti-CD150 or UPC-10 (isotype control) antibody for 48 h. Then, a flow cytometry analysis was used to find phenotypes representing activation, as described in the materials and methods. Cells were collected and incubated with FITC-conjugated anti-IL10, IgD, and CD80 Abs and PE-conjugated anti-CD38 Abs for the flow cytometry analysis. The thin-lines represent the results of staining with the negative control; dotted-lines represent the results of staining with the isotype control, and the thick-lines depict the expression of the indicated molecules induced by anti-CD150 Ab treatment. The results represent three independent experiments. UPC-10, isotype control; h., hours; anti-CD150 Ab, anti-CD150 antibody treatment.

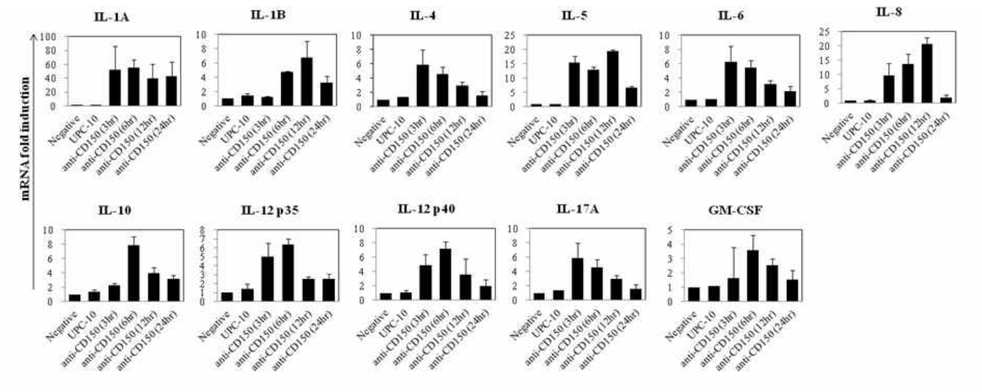

Figure 8. Effects of CD150 on inflammatory cytokine mRNA transcription. Quantitative real-time PCR was performed for various inflammatory cytokines 3, 6, 12, and 24 hours after CD150 stimulation with antibody cross-linking in EBV-transformed B cells. Data shown are as the mean ±SD of duplicate experiments. All results represent three independent experiments.

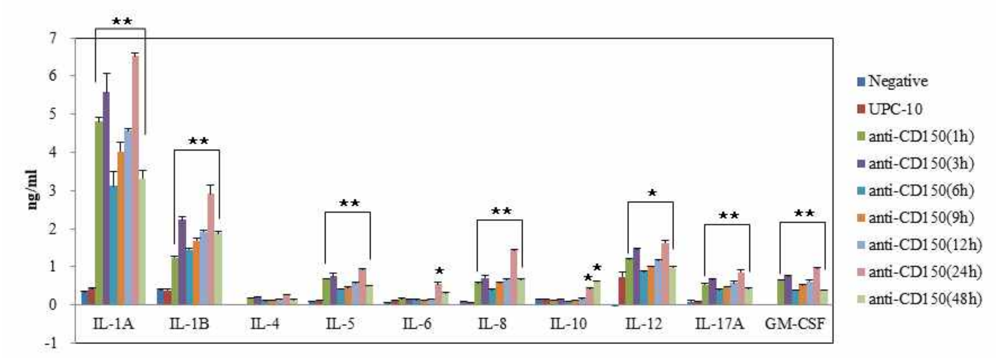

Figure 9. Production of multiple inflammatory cytokines after treatment with anti-CD150 Ab. EBV-t B cells were incubated with anti-CD150 or UPC-10 (isotype control) antibody for 48 h. Cell culture supernatants were collected after the indicated time (3-48 h), and the levels of various cytokines were measured using individual ELISA assay kits. Data are presented as the means of two independent experiments, and error bars represent ±SD of the means. Comparisons between individual data points were made using one-way ANOVA. *p <0.05 or **p<0.01, CD150-stimulated groups versus UPC-10 (isotype control) groups. h, hours; UPC-10, isotype control; anti-CD150, anti-CD150 antibodies treatment.

E. Src/PI3K signaling pathways were involved in CD150-induced EBV-t B cell response

Next, the regulatory signaling pathways leading to the CD150-induced expression of adhesion molecules and cytokine production were investigated. EBV-t B cells were incubated after CD150 stimulation using anti-CD150 Ab and cross-linking with secondary antibody. PI3K/Atk and Src were phosphorylated in EBV-transformed B cells following stimulation with anti-CD150 Ab (Fig. 10).

F. MAPK inhibitors blocked CD150-induced adhesion molecule expression and phenotype changes in EBV-t B cells

To confirm the involvement of the SRC/PI3K signaling pathway in CD150-induced EBV-t B cell responses, an inhibition assay was performed using mitogen-activated protein kinase (MAPK) inhibitors. The classical MAPK pathway associated with adhesion-molecule expression passes through SRC/ERK2 and SRC/JNK signaling [18, 19], so SB203580, a specific ERK2 inhibitor, and SP600125, a specific JNK inhibitor, were separately administered to CD150-stimulated EBV-t B cell. As expected, both inhibitors completely blocked the adhesion molecule expression induced by CD150 stimulation (Fig. 11). In the previous results, PI3K was activated by anti-CD150 Ab treatment. Therefore, LY294002, a specific PI3K inhibitor, was applied to CD150-stimulated EBV-t B cells. LY294002 effectively inhibited the phenotype changes (expression of IL-10, IgD, CD38, and CD80) induced by CD150 stimulation (Fig. 12).

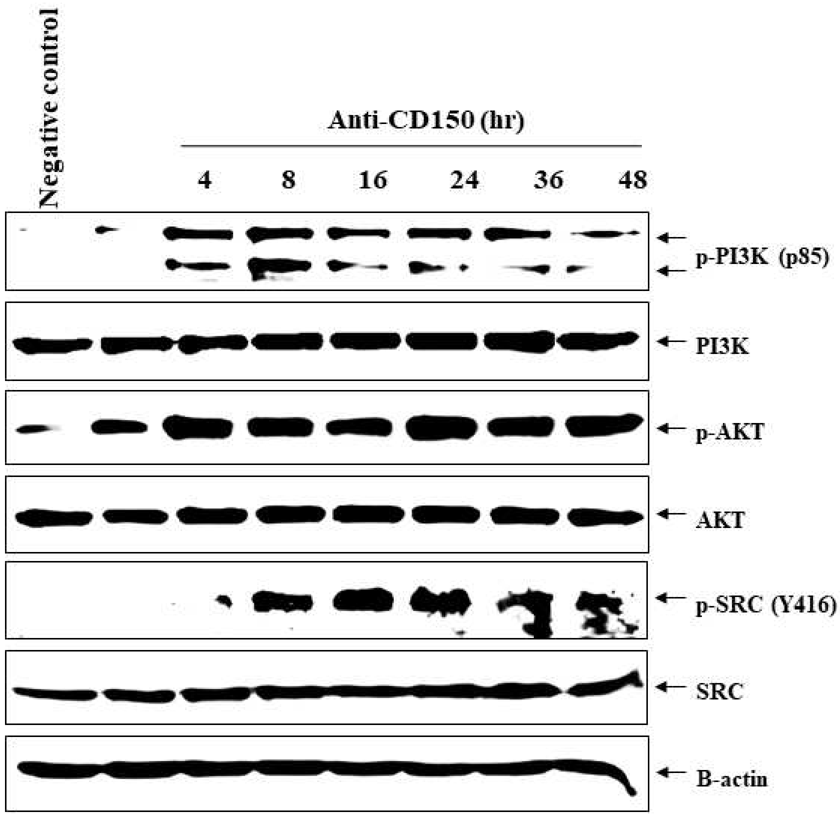

Figure 10. Intracellular signaling pathways involved in CD150-mediated EBV-t B cell responses. EBV-transformed B cells were incubated with anti-CD150 mAb (1 μg/ml, 30 min) or UPC-10 (1 μg/ml, 30 min) and then treated with anti-mouse IgG (1 μg/ml, 20 min). Cells were collected and further retained in cell culture medium for the estimated times, and then the total cell lysates were collected. All cell lysates were predisposed to immuno-blotting using anti-phospho-PI3K, -PI3K, phospho-AKT, -AKT, -phospho-SRC (Y416), -SRC, and - beta actin (loading control) antibodies. The results represent three independent experiments.

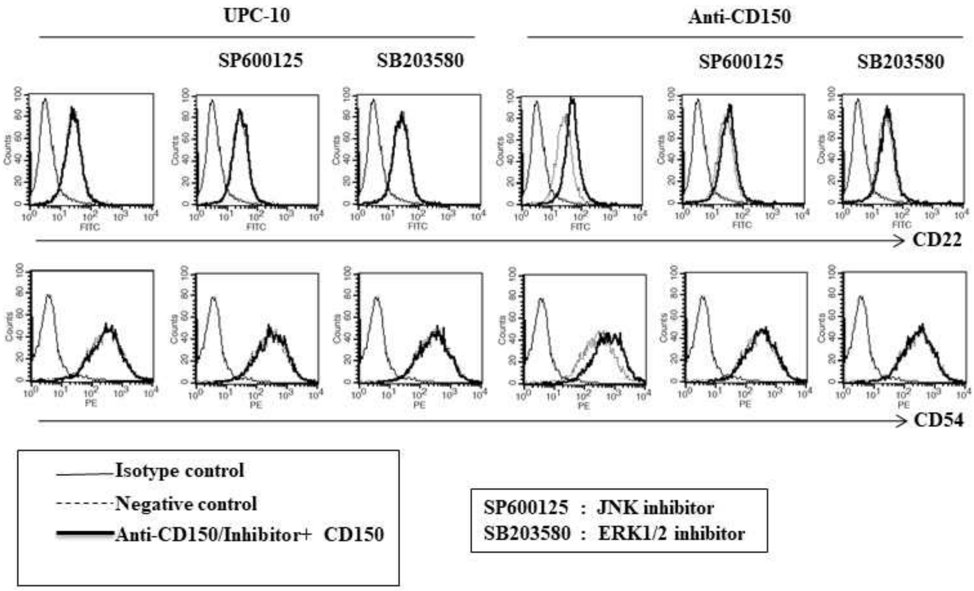

Figure 11. Flow cytometry analysis of adhesion molecules expression in CD150-stimulated EBV-t B cells after inhibitor treatment. CD150-stimulated EBV-transformed B cells pre-treated (or not) with SP600125, a specific JNK inhibitor; and SB203580, a specific ERK1/2 inhibitor. EBV-transformed B cells mock stimulated with UPC-10 Abs were used as the isotype control. Cells were collected and incubated with FITC-conjugated anti-CD22 and PE-conjugated anti-CD54 Abs for the flow cytometry analysis. The thin-lines represent the results of staining with the isotype control; the dotted-lines represent the results of staining with the negative control, and the thick-lines depict the expression of the indicated molecules induced by anti-CD150 Ab treatment with or without specific inhibitors. The results represent three independent experiments. UPC-10, isotype control; anti-CD150 Ab, anti-CD150 antibody treatment.

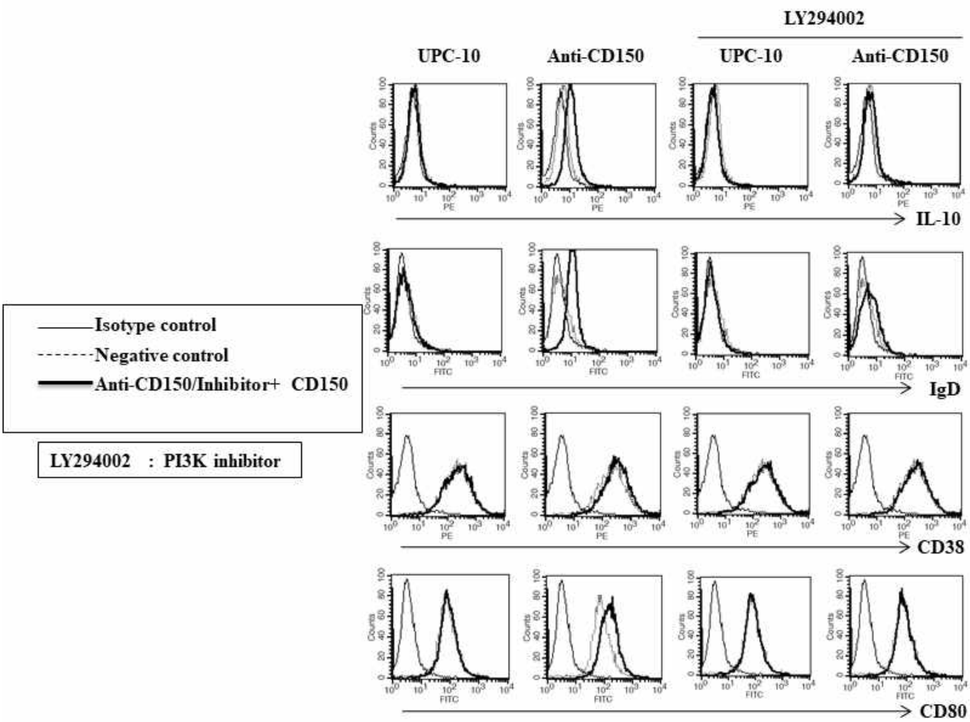

Figure 12. Flow cytometry analysis of phenotype changes in CD150-stimulated EBV-t B cells after inhibitor treatment. CD150-stimulated EBV-transformed B cells pre-treated (or not) with LY294002, a specific PI3K inhibitor. EBV-transformed B cells mock stimulated with an isotype control antibody (UPC-10) were used as the control. Cells were collected and incubated with FITC-conjugated anti-IL10, IgD, and CD80 Abs and PE-conjugated anti-CD38 Ab for the flow cytometry analysis. The thin-lines represent the results of staining with the isotype control; the dotted-lines represent the results of staining with the negative control, and the thick-lines depict the expression of the indicated molecules induced by anti-CD150 Ab treatment with the specific PI3K inhibitor. The results represent three independent experiments. UPC-10, isotype control; anti-CD150 Ab, anti-CD150 antibody treatment.

V. Discussion

CD150 is a pivotal signaling molecule for the production of various cytokines, and this study provides promising and novel data indicating that EBV-transformed B cells express of a variety of cytokine genes and secret multiple cytokines, which is likely to contribute to the activation of host immune responses. In humans, CD150 is constitutively expressed on immature thymocytes and $CD45RO^{+}$ memory T cells, and it is rapidly induced on naïve B and T cells following activation [20, 21]. CD150 is a self-ligand thought to play an important role in adhesion and signaling in the immune synapse between antigen-presenting cells and T cells [22]. Moreover, it has been suggested that CD150 could be involved in cancer malignancy and autoimmunity disorders including inflammatory bowel syndrome, rheumatoid arthritis, multiple sclerosis, hepatitis, systemic lupus erythematosus, and XLP [3, 4]. The data reported here show that CD150 was gradually expressed on the B cell surface during EBV infection and fully expressed on the surface after 3 weeks of EBV exposure (Fig. 1 and 2). Thus, CD150 might not only be induced on B cells immediately after EBV infection, but also increase in expression level over time.

Treatment with an anti-CD150 mAb was expected to increase apoptosis in EBV-transformed B cells. This expectation was based on the anti-proliferative characteristics of CD150, which inhibits proliferation in Hodgkin's lymphoma cell lines and induces cell death upon stimulation [23]. Interestingly, in this study, CD150 stimulation of EBV-transformed B cells under optimal conditions led to an increase in cell proliferation (Fig. 6), phenotype changes (Fig. 7), and the secretion of multiple cytokines, including pro-inflammatory cytokines (Fig. 8). Thus CD150 stimulation by antibody cross-linking on B cells infected with EBV might result in the production of multiple cytokines, which can affect immune cell proliferation and differentiation.

Although the expression of some genes (e.g. IL-10) [24] by EBV-associated lymphoblastoid cells has already been reported in the literature, the results presented here provide evidence that EBV-transformed B cells are potential producers of a variety of regulatory cytokines. Some of them can regulate EBV-infected B cell proliferation directly (IL-1, IL-10), and others are important regulators of leukocyte activity (IL-4, IL-5, IL-10) and the differentiation of peripheral blood monocytes (IL-4, IL-12, GM-CSF).

Actually, several viruses use CD150 receptors to maintain equilibrium with their host and evade immune surveillance. These viruses have developed a variety of immune evasive programs based on the regulation of cell surface molecules that serve as viral receptors and regulate cytokine expression [25, 26].

Finally, the regulatory signaling pathways leading to the CD150-induced expression of adhesion molecules and cytokine production were investigated. Based on previous studies, the involvement of the SRC/PI3K signaling pathway was proposed to be involved in the mechanism of the CD150-induced EBV-t B cell response [18, 19]. Western blot results and inhibition assays confirmed that the adhesion molecule expression passed through the SRC/ERK2 and SRC/JNK signaling, and the phenotype changes occurred through the PI3K/AKT signaling pathway (Figs. 11, 12, 13). These results also suggest that the CD150 receptor might be able to trigger the

SAP/Fyn and Src/PI3K signaling pathways to produce multiple cytokines in EBV-transformed B cells. CD150 binds to both SHP-2 and SHIP, and signaling through the CD150 receptor induces recruitment of SAP, (SH2D1A) to CD150 [27-29]. SAP serves an important role as an adaptor molecule that promotes the interaction between Fyn and the CD150 receptor and thereby facilitates receptor phosphorylation [30, 31]. Mice lacking SAP manifest increased IFN-ɣ secretion and reduced IgE levels, either at baseline or after treatment with several drugs, indicating that a deficiency in SAP expression results in a condition resembling a Th1 phenotype [32]. Furthermore, in SAP-deficient XLP patients, concentrations of IFN-ɣ are elevated during primary EBV infection [33], implying that a dysregulation in Th1 cytokine production could contribute to the progressive immunopathology observed in those patients.

In this study, I examined the effects and underlying mechanism of CD150 stimulation in EBV-transformed B cells. My findings, as summarized in Fig. 13, show that the increased CD150 expression induced by EBV infection effectively causes SRC/PI3K activation (phosphorylation) as an early event in the EBV-transformation of B cells. The activated SRC/PI3K provokes adhesion-molecule remodeling, cell proliferation, and cytokine production via the ERK/JNK and AKT activation signaling pathways (Fig. 13).

Collectively, these results provide novel insight into the mechanism of CD150-mediated EBV-t B cell responses including proliferation, cytokine production, and phenotype changes.

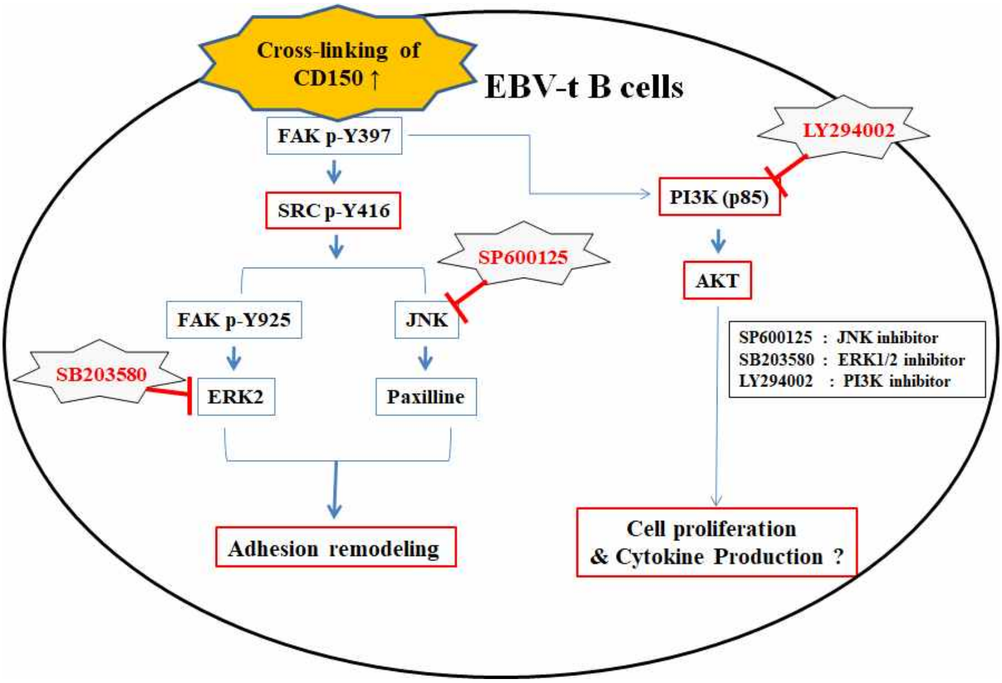

Figure 13. Schematic diagram of intracellular signaling mechanisms during the CD150-stimulated proliferative response in EBV-transformed B cells. Cross-linking of CD150 induced by EBV infection increases the expression of adhesion molecules through SRC/ERK2 and SRC/JNK phosphorylation signaling and leads to proliferation and the production of multiple cytokines through the PI3K/AKT activation signaling pathway. Three MAPK inhibitors effectively inhibited the CD150-stimulated proliferative response in EBV-transformed B cells. EBV-t B cells, EBV-transformed B cells.

Ⅵ. Conclusion

Taken as a whole, these results provide a novel insight into the mechanism of the CD150-stimulated EBV-t B cell response, and suggest that EBV-t B cells should be considered not only as targets for unusual regulatory cytokines but also as potentially active participants in local and systemic cytokine networks. More studies are needed to better understand the potential roles of CD150 in carcinogenesis and cytokine production in EBV-transformed B cells and EBV^+ lymphoma.

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